

Deliverable 1: Nodes, Topics, and Launch Files

Your name

Course / date

Purpose

This deliverable documents the ROS 2 nodes, topics, and launch behavior used in the two-drone scenario. The answers below should be based on observations from the launch file, ROS command-line tools, `rqt_graph`, and `rViz`.

Commands and Evidence Used

- Launch command tested:

```
ros2 launch two_drones_pkg two_drones.launch.yaml static:=True
```

- Node inspection commands used:

```
ros2 node list
ros2 node info /node_name
```

- Topic inspection commands used:

```
ros2 topic list
ros2 topic info /topic_name
ros2 topic echo /topic_name
```

- Visual tools checked: *Briefly note what you observed in `rqt_graph` and `rViz`.*

1 Nodes Running in the Static Two-Drone Scenario

How I Checked

I used `ros2 node list`.

Node List

- `/av1broadcaster`
- `/av2broadcaster`
- `/plots_publisher_node`
- `/rviz2_node`

- /transform_listener_impl_571acd970510
- /transform_listener_impl_5e7d7eb49390

2 Running the Scenario Without `ros2 launch`

Manual Commands

1. Terminal 1:

```
ros2 run tf2_ros static_transform_publisher 1 0 0 0 0 0 world av1
--ros-args -r __node:=av1broadcaster
```

2. Terminal 2:

```
ros2 run tf2_ros static_transform_publisher 0 0 1 0 0 0 world av2
--ros-args -r __node:=av2broadcaster
```

3. Terminal 3:

```
ros2 run two_drones_pkg plots_publisher_node
```

4. Terminal 4:

```
rviz
```

3 Published and Subscribed Topics

How I Checked

I used `ros2 node info <topic-name>`.

/av1broadcaster publishes the av1 frame on /tf_static.

/av2broadcaster publishes the av2 frame on /tf_static.

/plots_publisher_node publishes visualization markers on /visuals.

/rviz2_node subscribes to /visuals and displays the drone meshes.

Frame Publishers

Question. What nodes are responsible for publishing the `av1` and `av2` frames?

Answer. `av1broadcaster` and `av2broadcaster` nodes are responsible to publish transforms for `av1` and `av2`.

rViz Drone Mesh Topic

Question. Which topic causes rViz to plot the drone meshes?

Answer. rViz uses *visuals* topic to plot the drone meshes.

4 Effect of Omitting `static:=True`

Observation

```
if static is True:
    launch static_transform_publisher for av1
    launch static_transform_publisher for av2

unless static is True:
    launch frames_publisher_node
```

Launch File Explanation

`if.` = run this node only when the condition is true

`unless.` = run this node only when the condition is false